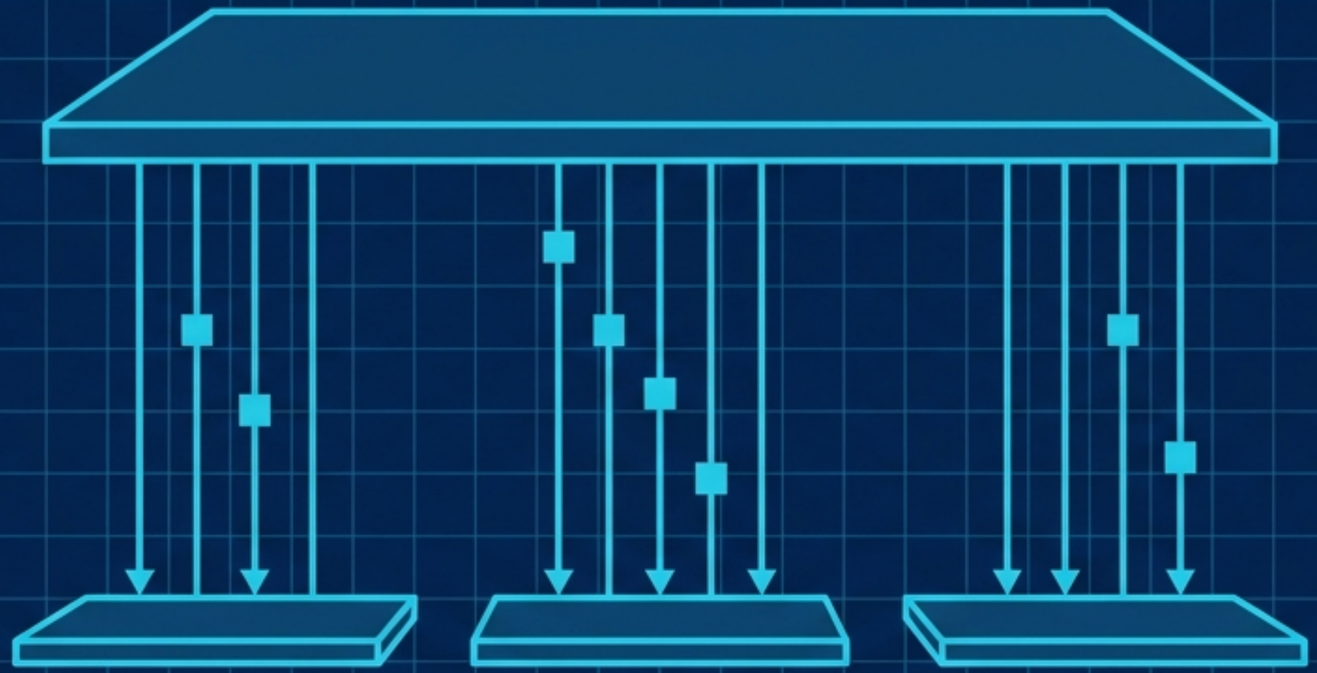


Domino Data Lab: Architectural Briefing & Placement Guide

Defending the Enterprise AI Platform in a Design Review

Document Status: Executive Decision Record | **Target:** Design Review Board

The Executive Ask & Strategic Stakes

The Recommendation (Change)

Deploy Domino Data Lab to mandate a governed, reproducible lifecycle (System of Record) across all data science workflows, standardizing how models move from IDE to App Hub.



The Stakes Sentence (Value)



If we do nothing, [Value at Risk] by [Date].



If we act, [Outcome] by [Date] at [Cost/Risk].

The Decision Today (Run)

Approve architecture mapping and capacity allocation.

Note: UNKNOWN parameters to be defined by project owner within 90 days.



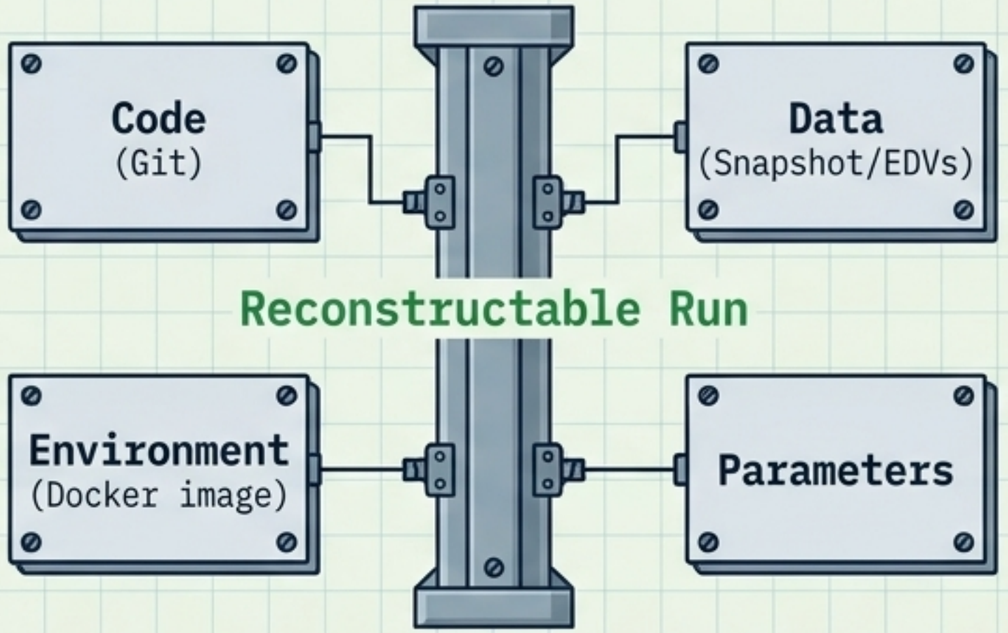
The Core Paradigm: System of Record over Hosted IDE

The Hosted IDE Trap



Files tracked, but environment and data are lost (Reproducibility Theater).

The System of Record

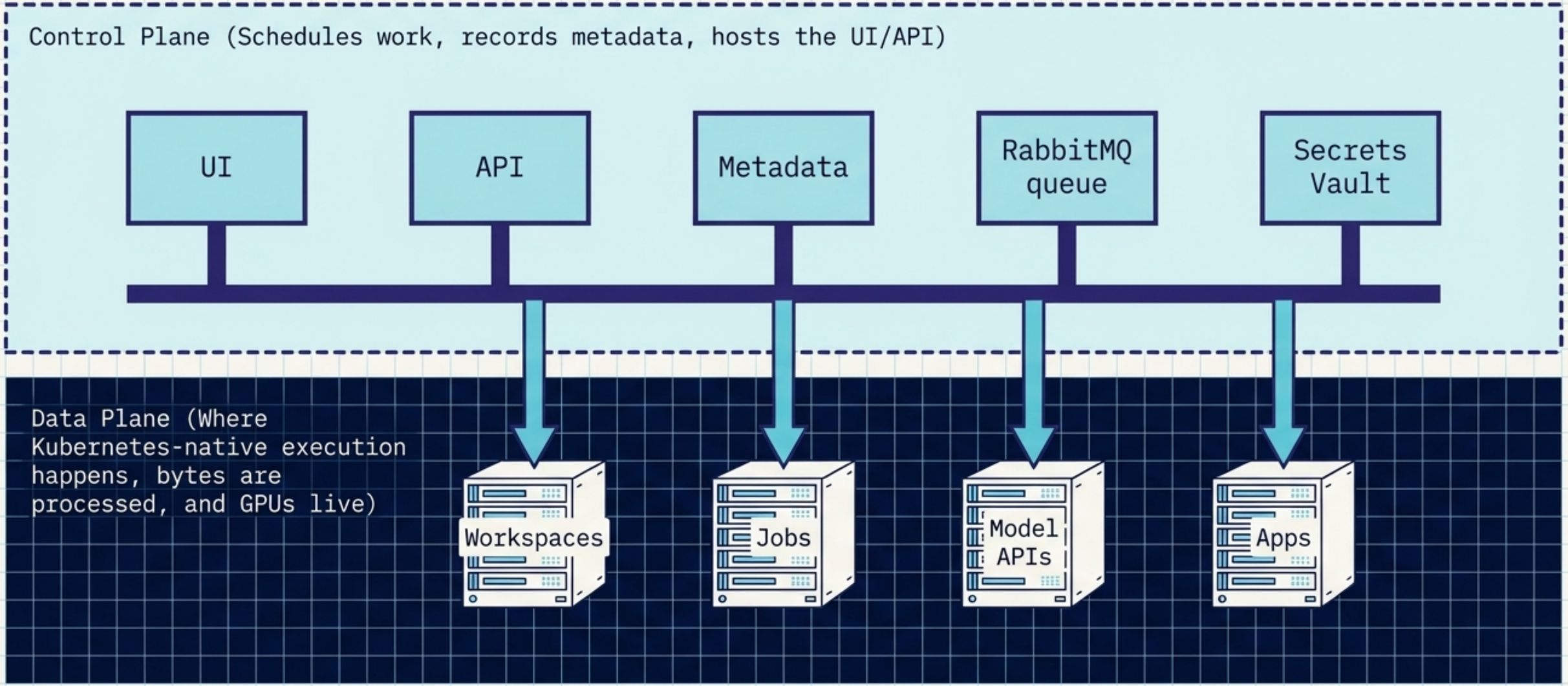


The platform's product is not the workspace; it is the cryptographic guarantee that a run can be reconstructed from zero.

Core Vocabulary: Mapping MLOps to Domino Primitives

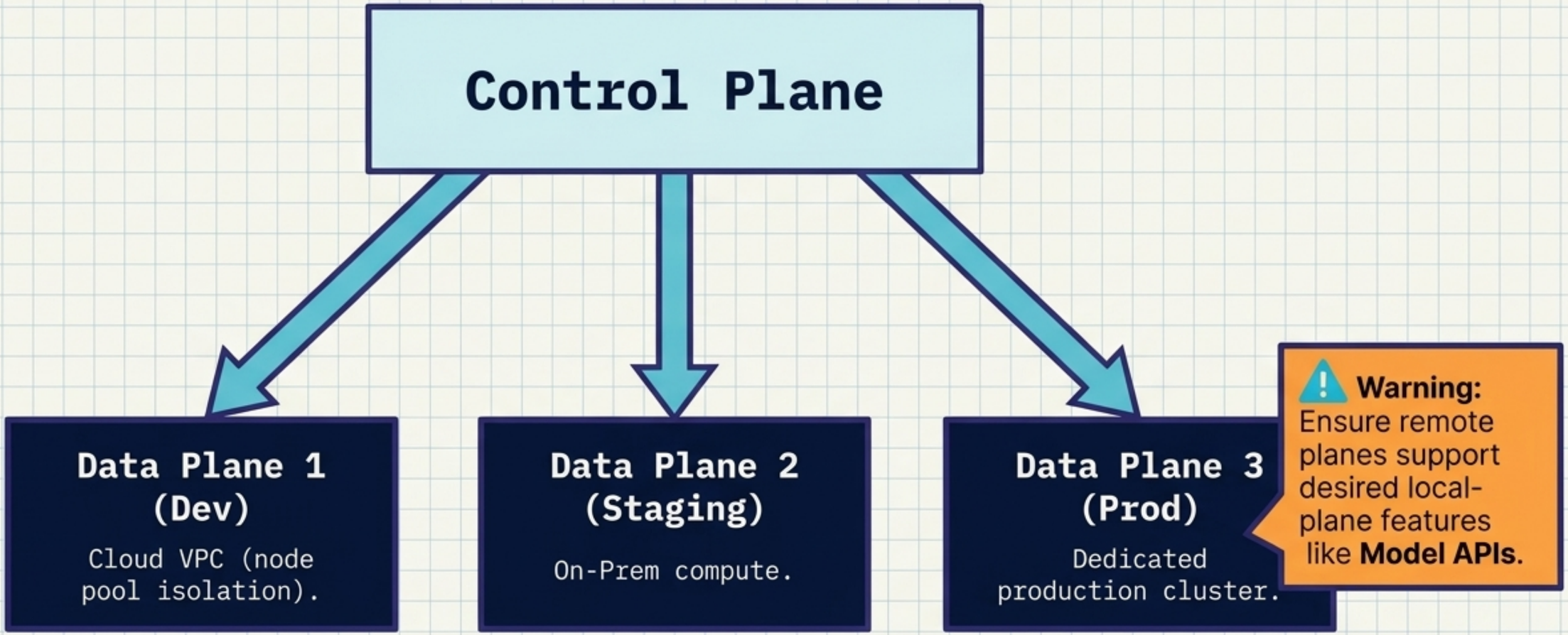
Generic MLOps Concept		Domino Primitive
Unit of Work		**Project** (The boundary for code, data, and access).
Hosted IDE		**Workspace** (Jupyter, VS Code, RStudio running on a specific hardware tier).
Container Image		**Environment** (Docker abstraction; e.g., Domino Standard Environment / DSE).
Serving / Endpoint		**Model API** (REST endpoint for machines).
Internal Tooling		**App Hub** (Streamlit/Dash UI hosted on port 8888 for humans).

Architectural Baseline: Separating Control from Compute



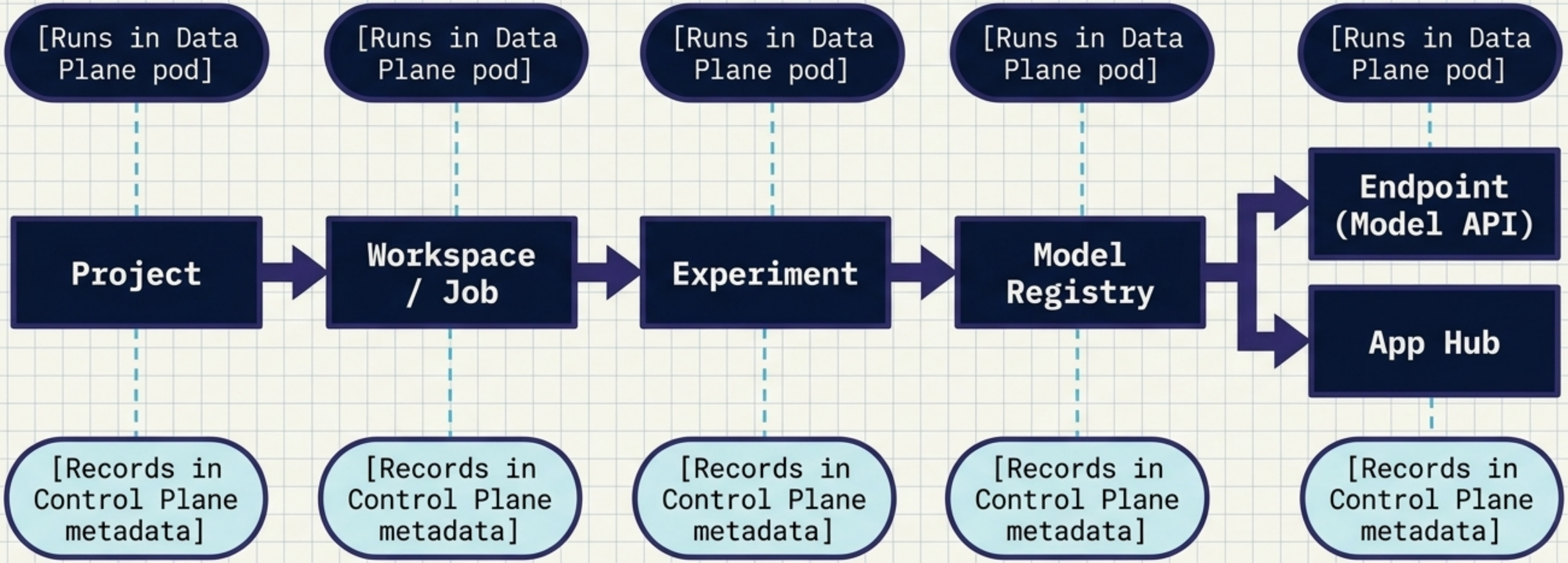
Domino Nexus:
One Control Plane, Many Data Planes

The Project story remains identical, but compute moves to where the data lives.

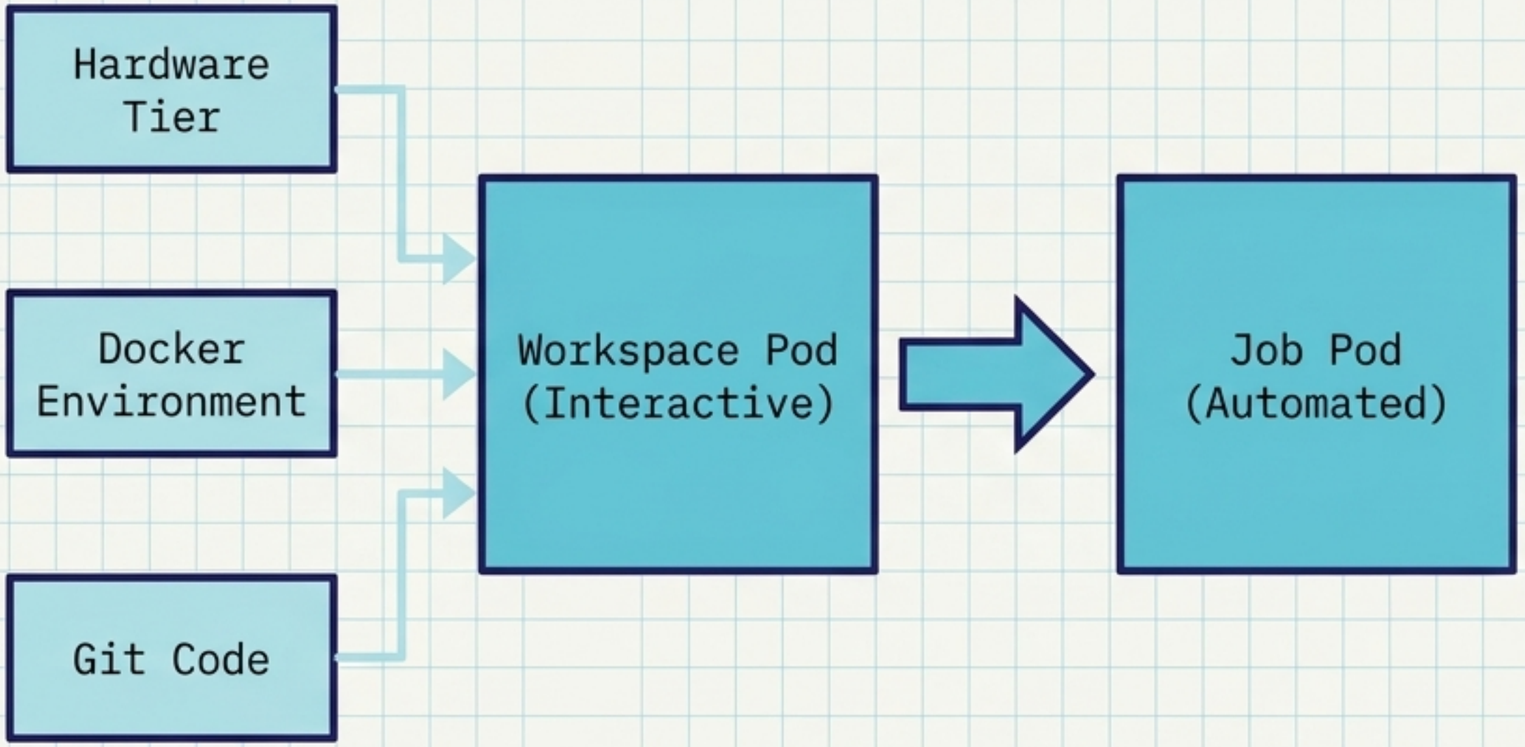


The Master Blueprint: The Governed Lifecycle

Every step in the Domino lifecycle is explicitly tracked in the Control Plane while executing in the Data Plane.



Phase 1: Build & Train (Interactive to Automated)



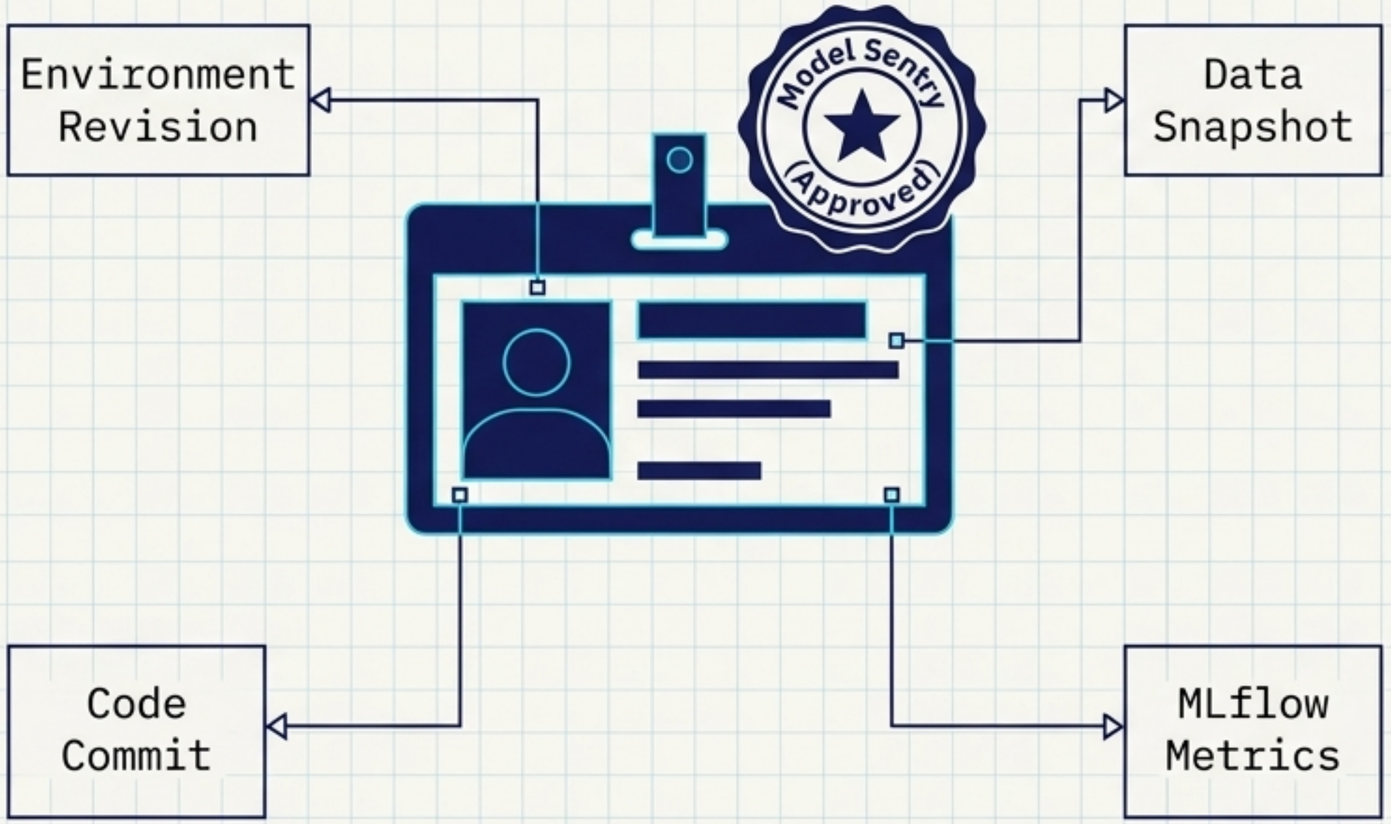
Workspaces:
Interactive development (Jupyter, VS Code, RStudio) powered by isolated node pools.

Environments:
Pre-configured Docker abstractions (DSE/DME) eliminate command-line dependency hell.

Jobs:
Transitioning to Jobs allows scheduled, headless execution for large-scale training and reproducibility.

Phase 2: Govern & Record (The System of Record)

The Anatomy of a Model Card

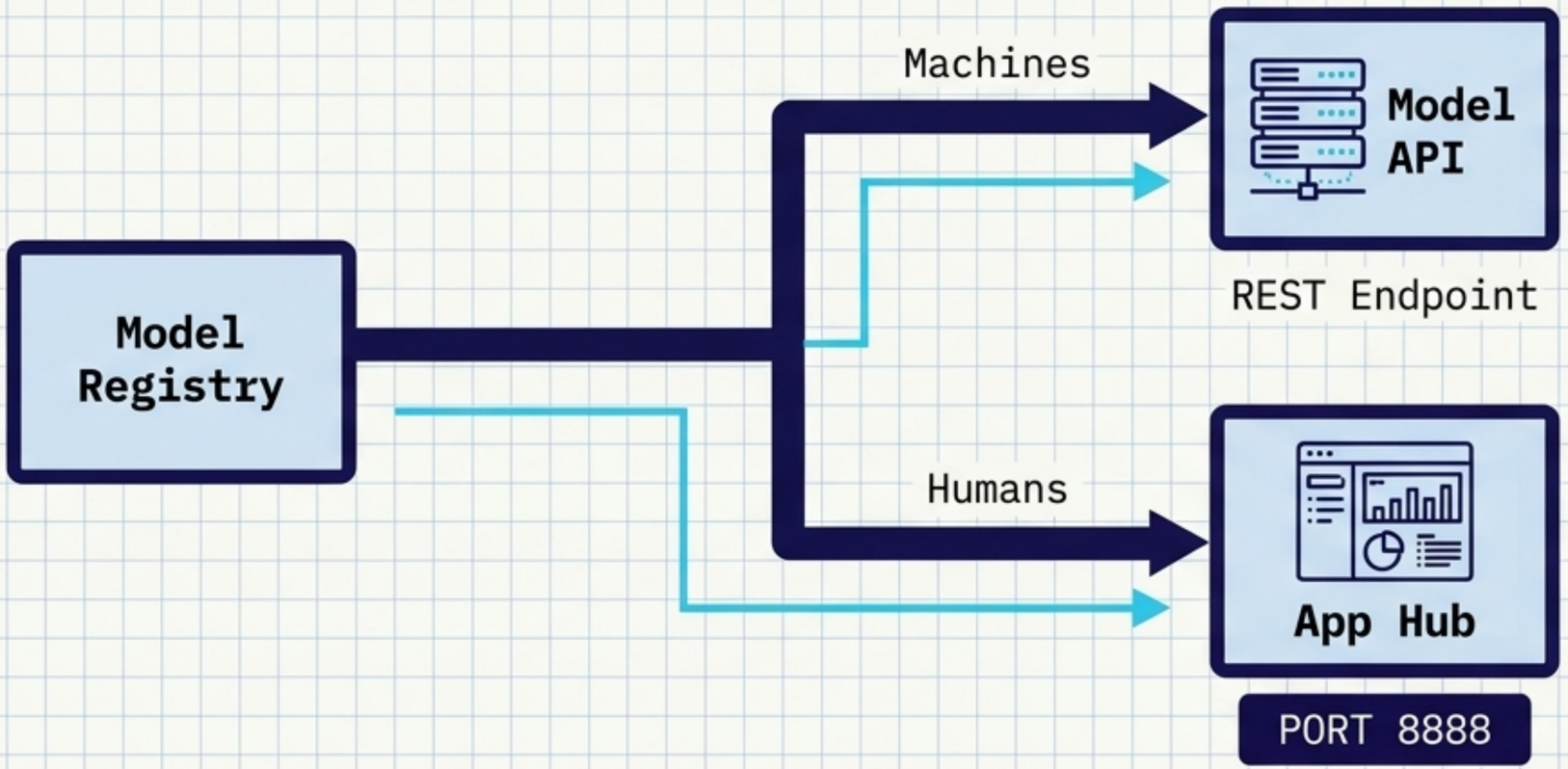


MLflow Integration:
Domino's registry wraps MLflow (per Chen et al. 2020) to capture standard experiment parameters.

The Model Card:
The ultimate artifact. Records exact code, Environment, Workspace settings, and datasets.

Model Sentry:
Enforces human-in-the-loop reviews (Dev → Staging → Prod) before deployment.

Phase 3: Serve & Scale (Deployment Options)



Serving Machines: Model APIs provide scalable, load-balanced REST endpoints hosted on Data Plane pods.

Serving Humans: App Hub scales interactive web applications (Streamlit, Dash) directly to business users.

Export Handoff

Exporting to SageMaker, Snowflake, or Fleet Command is supported, but this is a handoff, not a second home for the metadata.

The Placement Matrix: When to Refuse the Job

Pre-Existing Condition	Right Placement
The lakehouse (Unity Catalog) already owns the data and the registry.	**Databricks** (Domino Nexus would just be a redundant control plane).
The account is entirely AWS-native and endpoints already exist.	**SageMaker**
The enterprise estate runs on OpenShift and and the team wants to manage that stack.	**OpenShift AI**
Team builds custom CRDs and rejects commercial control planes.	**Kubeflow**
Primary users are analysts requiring a visual/no-code workbench.	**Dataiku**

Architectural Traps & Mitigation Ledger

The Trap	The Tell (Symptom)	The Fix (Mitigation)
Reproducibility Theater	Files are tracked in Git, but the Environment revision is missing.	Pin the exact Docker Environment and Data version.
Idle GPU Burn	A Workspace is left open on an expensive hardware tier.	Stop the Workspace; implement chargeback as a policy (not just a feature).
“Any Tool = No Lock-in”	“We can just export the model to SageMaker.”	Understand that while artifacts export, the System of Record metadata stays locked in Domino.

Compliance Realities: Separating Claims from Authorizations

Marketing Claims



The IL5 Myth: Homepage claims of “deployed in DoD IL5 environments” represent a program, not an Authorization to Operate (ATO).

Briefing Percentages: Claims of “6x ROI” or “40% infra savings” are briefing color.

Operational Reality



The Reality: A design review requires a primary DISA PA or FedRAMP package for a named offering, plus a local ATO.

The Metric: Measure your own queue to establish an SLA.

The Killer Questions (Design Review Defense)

Why now?	What if we wait?	Who else has done this?
Because ‘laptop research projects’ are creating un-auditable governance debt.	We compound the cost of reproducing lost models and idle GPU burn.	[UNKNOWN - Insert relevant analogue/benchmark before review].

What are we stopping?	What if this slips?
Ad-hoc container building and manual MLflow tracking.	We default to isolated shadow-IT Kubernetes clusters for data science.

Readiness Falsifier & Uncertainty Register



Readiness Falsifier

- ☐ Can you sketch the governed lifecycle from zero?
- ☐ Can you define Control Plane vs. Data Plane?
- ☐ Can you name one job Domino should refuse?

Uncertainty Register

Pending Facts	Owner	Date
Gap: DISA PA / FedRAMP package URL	Compliance Team	TBD
Gap: Remote Data-Plane feature parity on **installed Nexus version**	Platform Architecture	TBD

Bottom Line: All architectural baselines are sourced; specific hybrid compliance artifacts require vendor verification.